VINAY.T

LINUX SCENARIO BASED Q&A DAY 2

Q1. System is Very Slow — Commands Take Several Seconds to Execute

Scenario:
Basic commands like ls, cd, or top are taking 5–10 seconds or more to respond.

Possible Root Causes:
- The system is low on available RAM and is relying heavily on swap space.
- High CPU usage or disk I/O wait times may be contributing to the slowness.

Steps to Diagnose and Resolve:
1. Check memory usage: free -h
2. Check swap usage: swapon --show
3. Check CPU load and disk I/O: top

iostat -xz 1

Tip:
Temporarily disable swap with swapoff -a to see if performance improves — this helps confirm if swap is the issue.
Consider adding more RAM or optimizing/removing memory-intensive applications.

Q2. df -h Shows 0% Free Disk Space, but du -sh Shows Minimal Usage

Scenario:
The disk appears full (df -h shows 0% free), but du -sh reports very little actual disk usage. No large files are visible.

Root Cause:
One or more files have been deleted but are still held open by running processes. These files continue consuming disk space until the processes release them.

Solution Steps:
1. Identify processes holding deleted files: lsof | grep deleted
2. Restart the process or terminate it: kill -9 <PID>

Tip:
After clearing large log files, always restart related logging services to ensure the space is truly freed.

Q3. High Load Average but Low CPU Usage

Scenario:
The top command shows a high load average (e.g., >5), yet the CPU remains mostly idle.

Root Cause:
Load average includes not only processes using the CPU but also those waiting for disk or I/O operations. A high load can result from I/O bottlenecks, not just CPU saturation.

Solution Steps:
1. Check I/O wait and disk performance: iostat -xz 1
2. Check for blocked or waiting processes: vmstat 1

Tip:
For I/O-intensive workloads, consider using SSDs and optimizing disk schedulers (e.g., noop or deadline) to reduce wait times.

Q4. Sudden Disk I/O Spikes — Application Performance Degrades

Scenario:
Applications experience random slowdowns. iostat reports high %util and await times, indicating disk bottlenecks.

Root Cause:
Excessive read/write operations from one or more processes are overwhelming the disk, causing latency for other applications.

Solution Steps:
1. Identify processes causing high I/O: sudo iotop
2. Pause or throttle the offending process: ionice -c2 -n7 -p <PID>

Tip:
Isolate disk-heavy workloads by using dedicated partitions or disks for logs, databases, or other I/O-intensive applications.

Q5. top Shows One Process Using 99% CPU

Scenario:
The system is sluggish, and top reveals a single process consuming nearly 100% of the CPU.

Root Cause:
This is often caused by a runaway script, infinite loop, or poorly optimized query consuming excessive CPU resources.

Solution Steps:
1. Identify the high-CPU process:
```
top
ps aux --sort=-%cpu | head
```
2. Pause or terminate the process:
```
kill -STOP <PID>     # To pause
kill -9 <PID>        # To forcefully kill
```

Tip:
For non-critical, CPU-intensive processes, use nice or cpulimit to limit CPU usage without fully stopping them.

Q6. Memory Usage Gradually Increases Over Time

Scenario:
RAM usage steadily climbs day by day, eventually leading to swap usage and degraded performance.

Root Cause:
This behavior typically indicates a memory leak in a long-running service, background job, or cron task that doesn't release memory properly.

Solution Steps:

1. Monitor memory usage and identify the culprit:
   top
   ps aux --sort=-%mem
2. Restart the leaking service on a schedule or as needed.

Tip:
Implement monitoring with tools like Prometheus + Grafana to visualize memory usage trends and catch leaks early.

Q7. Disk Write Speed is Abnormally Low

Scenario:
Copying large files is taking much longer than expected, even on seemingly idle systems.

Possible Root Causes:

- Disk I/O contention from other processes
- Inefficient or inappropriate disk scheduler
- Disk fragmentation (mainly on spinning drives)
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Solution Steps:

1. Benchmark write speed:
   dd if=/dev/zero of=testfile bs=1G count=1 oflag=dsync
2. Check current I/O scheduler:
   cat /sys/block/sdX/queue/scheduler

Tip:
Use the deadline or noop scheduler for SSDs to optimize write performance.

Q8. CPU Constantly at 100% Across All Cores

Scenario:
The system shows high CPU usage on all cores, even when seemingly idle.

Root Cause:
A background process, such as a misconfigured cron job, runaway rsync, or even malware, may be consuming CPU continuously.

Solution Steps:

1. Inspect running processes:
   top
   htop
   ps aux --sort=-%cpu
2. Identify and kill the offending process, or investigate further if suspicious.

Tip:
Use tools like auditd or psacct to monitor and log system activity for detecting rogue or abnormal behavior.

Q9. RAM Appears Full, but System Performance is Normal

Scenario:
The free -h command shows high memory usage, yet the system remains responsive with no noticeable lag.

Root Cause:
Linux uses free memory to cache filesystem I/O, improving performance. This cached memory is released when needed by applications.

Solution Steps:
1. Use the following to interpret memory usage correctly: free -h
2. Focus on the "available" column, not just the "used" memory.

Tip:
High memory usage in Linux is normal. Cached memory is not wasted; it's part of the system's optimization strategy.

Q10. Application Slows Down During File Copy Operations

Scenario:
Application performance degrades noticeably whenever large files are being copied on the system.

Root Cause:
Disk I/O contention: both the application and the file copy operation are competing for the same disk resources.

Solution Steps:
1. Run copy operations with low I/O priority: ionice -c3 cp file /target/

Tip:
Isolate high-I/O operations by using separate storage for applications, backups, and logs to prevent performance bottlenecks.